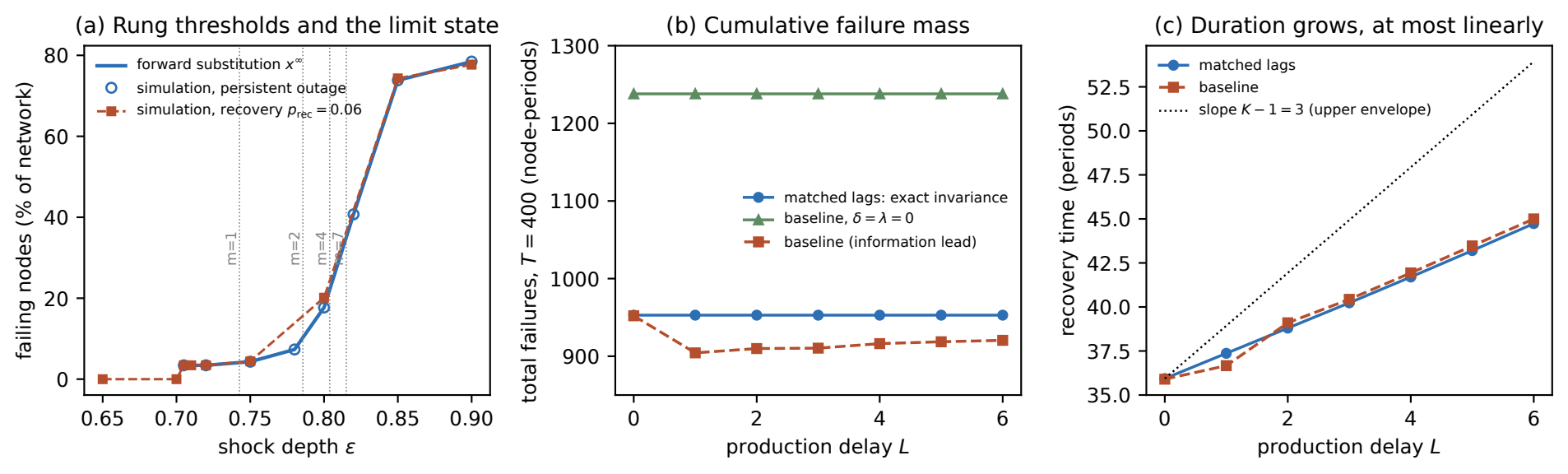
(a) Rung thresholds and the limit state
failing nodes (% of network)
shock depth ε
forward substitution x∞
simulation, persistent outage
simulation, recovery p_rec = 0.06
m=1
m=2
m=4
m=7
(b) Cumulative failure mass
total failures, T = 400 (node-periods)
production delay L
matched lags: exact invariance
baseline, δ = λ = 0
baseline (information lead)
(c) Duration grows, at most linearly
recovery time (periods)
production delay L
matched lags
baseline
slope K − 1 = 3 (upper envelope)